High school · reasoning

Parity and invariants

Synthesis problem

Investigate, justify, conclude

An 8×8 chessboard is coloured alternately black and white. Remove two diagonally opposite corners. Can the 62 remaining squares be tiled by 31 dominoes of size 1×2, each covering two edge-adjacent squares? Count the removed colours and those covered by a domino. Compare with removing two neighbouring corners.